

# CrimeGraph AI

*An Explainable AI-Powered Criminal Network Intelligence & Investigation Platform*

**DETAILED PROJECT REPORT**

|                         |                                                             |
|-------------------------|-------------------------------------------------------------|
| Problem Statement Title | AI-Powered Criminal Network Analysis System                 |
| Organization            | Ministry of Home Affairs                                    |
| Department              | National Crime Records Bureau (NCRB), Women Safety Division |
| Category                | Software                                                    |
| Theme                   | Blockchain & Cybersecurity                                  |
| Proposed Solution Name  | CrimeGraph AI                                               |

## 1. Executive Summary

Most investigating agencies don't really have a shortage of evidence. FIRs, call detail records, IPDRs, bank ledgers, UPI trails, vehicle logs, prior-conviction data — all of it usually already exists somewhere. The actual problem is that it's scattered across a dozen unconnected systems, and nobody has the time to manually check whether a phone number in one case file matches a phone number in another case filed three states away. CrimeGraph AI is basically our attempt at closing that gap: a platform that pulls this scattered evidence into one connected view and lets an investigator ask, in plain language, how two people, cases or transactions are actually related.

The core idea is that the system takes every person, phone, bank account, vehicle, case and location an officer already has on record, and instead of storing them as rows in separate tables, represents them as one connected network. Once that network exists, you can run graph algorithms on it (the same basic idea used to rank web pages or catch fraud rings in banking) to find things a manual case review would almost certainly miss. Things like an organiser who barely shows up in call records but still sits at the centre of three separate groups, an account that briefly bridges two gangs that otherwise have nothing to do with each other, or two FIRs from different police stations that happen to share a phone number nobody had reason to compare.

One thing we tried to be careful about throughout the design: this is supposed to help an investigator, not replace their judgement. Every result the system gives comes with the actual source records and a confidence score attached, so an officer can go check the reasoning himself instead of just trusting a number on the screen.

## 2. Understanding the Problem

### 2.1 Why this keeps happening

Organised crime groups don't run things to make life easy for investigators. SIM cards get swapped every few weeks, money gets routed through mule accounts opened for a single transaction, and the person actually giving the orders usually stays nowhere near the crime scene. This isn't accidental, it's built that way on purpose, to keep some distance between whoever is running the operation and whoever eventually gets caught. On the enforcement side, the evidence that could expose these layers is spread across agencies and formats that were never designed to be compared against each other, so it basically comes down to an officer's memory or luck to catch a connection.

- Data silos: FIR records, telecom dumps and bank transaction data usually sit in separate departmental systems, often in different states altogether.
- Inconsistent formats: a single case can involve handwritten or scanned regional-language FIRs along with tabular CDRs and structured transaction logs, and none of these open in the same tool.
- Sheer volume: one multi-accused case alone can generate lakhs of CDR rows and thousands of transactions, which is just more than any team can realistically go through line by line.
- Deliberate distance: a layered setup exists specifically to keep the person directing the syndicate two or three hops away from anything an officer would normally think to search for.

### 2.2 Where the tools currently in use fall short

We're not the first ones trying to solve pieces of this problem, so it's only fair to be upfront about what already exists instead of acting like this space is empty.

| Tool / Approach                                         | What it does well                                                                                          | Where it falls short for this problem                                                                                                               |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| CCTNS / state case-management systems                   | Standardised FIR digitisation and case tracking across most states.                                        | Built for record-keeping, not cross-case relationship discovery; no graph or network view.                                                          |
| Standalone CDR-analysis software                        | Good at visualising call patterns for a single number or case.                                             | Doesn't connect to financial, vehicle or FIR data; each case is analysed in isolation.                                                              |
| IBM i2 Analyst's Notebook / similar link-analysis tools | Mature link-charting and manual network visualisation, used by several intelligence units internationally. | Largely manual entity linking, licensing cost, and limited support for Indian-format data (regional scripts, local telecom formats) out of the box. |
| Manual cross-referencing (current default)              | Flexible, requires no new tooling or training.                                                             | Doesn't scale past a handful of suspects; entirely dependent on an individual officer noticing a match.                                             |

Honestly, we're not really inventing a new category of technology here. Graph databases, NLP-based document processing and financial-anomaly detection are each fairly mature on their own already. What's missing is one platform that runs all three on the same connected dataset, built specifically around the kind of data Indian investigating agencies actually work with.

### 3. Proposed Solution

The basic approach is to give an investigator one starting point, whether that's a name, a phone number or an FIR number, and let the system build the surrounding network automatically, instead of making the officer piece it together by hand across five different tools.

#### 3.1 Core workflow





### 3.2 A worked example

Maybe it's easier to just walk through a made-up example to show why the graph actually helps. Say an officer already has these four records, sitting in four completely different systems:

| Source         | Record                                                                         | System it lives in           |
|----------------|--------------------------------------------------------------------------------|------------------------------|
| FIR-1042       | Person A named as an accused                                                   | State case-management system |
| CDR dump       | Person A's phone called Person B fourteen times over three days                | Telecom CDR extract          |
| Bank statement | Person B's account transferred ₹40,000 to Person C in five split payments      | Financial intelligence unit  |
| Vehicle log    | A vehicle registered to Person C was recorded near the scene named in FIR-1042 | Traffic/ANPR database        |

On their own, none of these four records looks unusual, and none of them would get flagged on its own. Whoever's checking the vehicle log has no particular reason to also be looking at the bank statement. But once you load all four into the graph, they line up into a single four-hop path from the named accused all the way to a vehicle placed at the scene, and the split-payment pattern between B and C is exactly what the mule-account detector is meant to pick up on. That's really the whole point in one example: it's not some new type of evidence, it's just the connection that nobody had a reason to go looking for manually.

## 4. Unique Features

### Cross-Jurisdictional Case Linker

This checks every new FIR against the existing graph for shared phone numbers, repeated aliases or common bank accounts, no matter which police station or state filed it. Right now this kind of overlap is only found when an officer happens to remember a name from some unrelated case.

### Dynamic Mastermind Identifier

This uses eigenvector and betweenness centrality (standard graph-theory measures, nothing custom-built) to flag people who have very few connections but still sit at important points in the network. Few contacts plus high structural importance is a decent proxy for someone who's directing things while staying personally quiet.

### **Smurfing and Mule-Account Detection**

Looks out for split-amount fund routing, sudden multi-hop transfers, and newly opened accounts that cluster around unusually active communication circles. These are fairly well-known patterns in money-laundering cases, so flagging them automatically saves a lot of manual work tracing a transaction chain by hand.

### **Multi-Dialect OCR and NLP Processing**

Reads scanned regional-language FIRs directly and pulls out names, locations, dates and stated relationships straight into the graph, so nobody has to retype a handwritten case narrative before it becomes searchable.

### **Tamper-Evident Chain of Custody**

Every query, export and edit on the graph gets logged in a way that can't just be quietly changed later. This is what keeps the analysis usable as supporting material in court, since an unverifiable data trail would otherwise be a real problem.

## **5. Hackathon Scope vs. Full Product Vision**

It wouldn't be fair to present all of this as something we can build end-to-end within a hackathon timeframe, so we've split the plan into what we can actually demonstrate right now and what the full platform looks like further down the line.

### **5.1 What we are building for this hackathon**

- A working ingestion pipeline for CSV/JSON case data (mock FIR, CDR and transaction datasets standing in for live agency feeds).
- Entity resolution across at least name, phone number and account fields, with a confidence score shown to the user.
- A Neo4j-backed knowledge graph populated from that pipeline, queryable through a Cytoscape.js visual interface.
- Multi-hop path search between two entities, and basic centrality scoring to highlight structurally important nodes.
- A simple natural-language query box that translates a handful of question patterns ("show connections of X within N hops") into graph queries.

### **5.2 What sits in the full product roadmap, not the prototype**

- Production-grade OCR across multiple regional scripts, requiring language-specific model tuning.
- Live integration with actual telecom and banking systems, which involves data-sharing agreements and compliance review well beyond a hackathon's scope.
- Court-admissibility certification of the chain-of-custody logging, which would need legal and departmental sign-off.
- Full role-based deployment across multiple states with departmental access-control policies.

We're keeping this split explicit on purpose. A demo that actually works on a realistic, well-defined scope is a lot more convincing than a slide full of features that were never really built.

## **6. System Architecture**

| Layer             | Function                                                                                                                                                              |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Ingestion    | Accepts CSV, JSON, Excel, relational DB exports and API feeds from authorised case-management and intelligence systems.                                               |
| Data Processing   | Cleans and standardises inconsistent phone formats, addresses and dates into canonical representations.                                                               |
| Entity Resolution | NLP and ML-based matching to determine whether records such as 'Rahul Kumar' and 'R. Kumar, 9876543210' refer to the same person, with a reviewable confidence score. |
| Knowledge Graph   | Neo4j graph database storing people, phones, vehicles, accounts, cases, locations and organisations as nodes, and their interactions as relationships.                |
| Graph Engine      | Centrality, community detection and multi-hop path analysis to surface structurally important entities and hidden connections.                                        |
| AI Engine         | Natural-language query translation, anomaly detection and relationship summarisation.                                                                                 |
| GIS Engine        | Spatial correlation of entities, events and movement patterns.                                                                                                        |
| Investigation UI  | Unified dashboard combining the network graph, timeline, map, case explorer and AI assistant.                                                                         |

## 7. Technology Stack

|          |                                                                        |
|----------|------------------------------------------------------------------------|
| Frontend | React, Next.js, Tailwind CSS, Cytoscape.js / D3.js for graph rendering |
| Backend  | Python, FastAPI                                                        |
| AI / ML  | PyTorch, scikit-learn, spaCy, transformer-based NLP models             |

|                            |                                                               |
|----------------------------|---------------------------------------------------------------|
|                            |                                                               |
| <b>Graph Database</b>      | Neo4j                                                         |
| <b>Relational Database</b> | PostgreSQL — users, case metadata, configuration, audit logs  |
| <b>Search</b>              | OpenSearch / Elasticsearch                                    |
| <b>GIS</b>                 | MapLibre, PostGIS                                             |
| <b>Caching / Queues</b>    | Redis                                                         |
| <b>Deployment</b>          | Docker on Linux, with GPU-enabled nodes for NLP/OCR workloads |

We went with a hybrid database setup on purpose instead of one single store. PostgreSQL handles the structured, transactional stuff like users and audit trails, while Neo4j is kept aside specifically for the relationship-heavy entities — people, phones, accounts, cases — where graph traversal is just faster than relational joins once you go past two or three hops deep.

## 8. Explainable AI: A Non-Negotiable Design Principle

The platform never just spits out a bare risk score. Something like '92% criminal probability' is basically useless in court and can't really be defended as good investigative practice anyway. So every insight the system gives comes with the supporting evidence behind it:

|                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Entity: Person C — Network role: Potential intermediary — Confidence: 86%</b></p> <p>Why flagged: connected to Person A through Phone X · linked to Account Y · appears in 3 relevant events · bridges two previously separate clusters</p> <p>Supporting records: CDR-102, TXN-341, CASE-208</p> <p>Recommended action: Review by investigating officer</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

This way, every AI output becomes a lead the officer can go check for himself, not a verdict he's just expected to trust.

## 9. Security & Privacy Architecture

## 9.1 Access control

Role-based access control runs through the whole system, split across four tiers: Administrator, Senior Investigator, Investigator and Analyst, and each one only gets the capabilities their role actually needs.

## 9.2 Data protection

- Encryption in transit and at rest for all case data.
- Secure credential management, no access keys stored in plaintext anywhere.
- Full audit logging for every access, export and graph edit — who did it, what changed, and when it happened.

## 9.3 Ethical guardrails

- Human-in-the-loop by design: the system only generates leads, the investigators make the actual decisions.
- No automated guilt labelling. Every output is phrased as a relationship worth reviewing, never as an accusation.
- Full evidence traceability, so every insight can be traced back to its source record.
- Data retention that follows whatever law and departmental policy applies.

# 10. Feasibility and Viability

## 10.1 Technical feasibility

None of the individual pieces here are experimental, to be clear. Graph databases, NLP-based entity extraction, OCR for regional scripts, centrality-based network analysis — all of this is production-grade tech already being used in other industries. The actual engineering effort is in the integration: building an entity-resolution pipeline that can hold up against messy, inconsistently formatted Indian administrative and telecom data, and keeping multi-hop graph queries fast once the dataset gets to a real investigative scale.

## 10.2 Key risks and mitigation

| Risk                              | Mitigation                                                                                             |
|-----------------------------------|--------------------------------------------------------------------------------------------------------|
| False-positive entity matches     | Confidence-scored matches are surfaced for officer confirmation rather than auto-merged.               |
| Data sensitivity / misuse         | Role-based access, full audit trails and encryption at every layer.                                    |
| Integration across legacy formats | Modular ingestion adapters per source type (FIR/CDR/financial/vehicle) rather than one rigid pipeline. |
| Query performance at scale        | Graph-native storage (Neo4j) with indexed multi-hop traversal, purpose-built for this access pattern.  |



## 11. Impact and Benefits

### 11.1 For investigators

- Cuts down hours of manual cross-referencing into one searchable network view.
- Surfaces indirect, multi-hop links that are basically invisible during manual review.
- Generates court-ready dossiers in one click, backed by traceable source records.

### 11.2 For the department

- Gives a shared intelligence layer across jurisdictions, so less investigative effort gets duplicated.
- Helps identify the actual organisers faster, not just the operational proxies working under them.
- Structured audit trails that make legal admissibility easier, not harder.

## 12. Implementation Roadmap

Past the hackathon prototype, a realistic path to something deployable would need to happen in phases rather than one big build. Here's our rough working estimate, based on the scope from Section 5.

| Phase   | Duration   | Focus                                                                                                                                               |
|---------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1 | 0–2 months | Harden the ingestion and entity-resolution pipeline against real-world data quality issues; pilot with anonymised or synthetic case data.           |
| Phase 2 | 2–5 months | Build out graph analytics (centrality, community detection, anomaly flags) and the GIS/timeline modules; internal testing with a single department. |
| Phase 3 | 5–9 months | Add regional-language OCR coverage, expand the natural-language assistant, and begin controlled pilot deployment with one investigating unit.       |
| Phase 4 | 9+ months  | Security audit, chain-of-custody legal review, and phased rollout across additional departments or states.                                          |

This is a plan, not a promise. The actual timelines would depend on data-access approvals and the department's own review cycles, and both of those are completely outside our control at this stage.

## 13. References

The concepts and techniques we've referenced in this report are all based on established, publicly documented work, not on anything proprietary:

- Neo4j Graph Data Platform documentation, for graph storage and Cypher query patterns.
- Freeman, L.C., "A Set of Measures of Centrality Based on Betweenness" — the basis for the betweenness-centrality approach used to identify structurally important entities.
- IBM i2 Analyst's Notebook product documentation, referenced for comparison in Section 2.2.
- Publicly available material on the Crime and Criminal Tracking Network & Systems (CCTNS), referenced for comparison in Section 2.2.
- spaCy and standard transformer-based NLP model documentation, for named-entity recognition on unstructured case text.

## 14. Conclusion

The problem the Ministry of Home Affairs has laid out isn't really about a lack of data. It's more a lack of synthesis. CrimeGraph AI is built around that exact distinction — it's not trying to replace investigative judgement with some black-box score, it's trying to give that judgement a complete, connected, evidence-backed picture to actually work from. By pulling FIRs, telecom records, financial trails and case data into one explainable knowledge graph, the platform is meant to do what manual analysis just can't: expose the organisers who stay hidden behind layers of operational proxies, while still keeping every step of that reasoning visible and defensible in court.